



AFFILIATED TO TRIBHUVAN UNIVERSITY

> BCA **> BBM**

Email: info@asm.edu.np

1. An airplane is moving in a circular path with speed 200 km/hr . The change in velocity in half revolution is
 - a. 0
 - b. 400 km/hr
 - c. 200 km/hr
 - d. $200\sqrt{2} \text{ km/hr}$
2. A smooth inclined plane is inclined at an angle θ with horizontal. A body starts from rest and sliding down the inclined plane, the time taken by body to reach the bottom is
 - a. $\sqrt{\frac{2h}{g}}$
 - b. $\sqrt{\frac{2l}{g}}$
 - c. $\frac{1}{\sin \theta} \sqrt{\frac{2h}{g}}$
 - d. $\sin \theta \sqrt{\frac{2h}{g}}$
3. A ball is thrown at angle 30° above horizontal from top of the tower with 10 m/s . The ball hit the ground at 17.3 m from foot of tower then height of tower is a.
 - a. 110 m
 - b. 100 m
 - c. 10 m
 - d. 12 m
4. A tachometer is device used to measure the
 - a. Tension on string
 - b. surface Tension
 - c. speed of rotation
 - d. centripetal force
5. A body is in SHM of time T . The time taken by it to move from extreme position to half of amplitude is
 - a. $\frac{T}{2}$
 - b. $\frac{T}{4}$
 - c. $\frac{T}{6}$
 - d. $\frac{T}{12}$
6. On heating liquid of cubical expansion α in a container having linear expansivity $\frac{\alpha}{3}$ the level of liquid in container will be
 - a. Rise
 - b. fall
 - c. will remain almost stationary
 - d. none
7. The total translational K-E of 5 moles of gas at 127°C is
 - a. $1.2 \times 10^4 \text{ J}$
 - b. $2.49 \times 10^4 \text{ J}$
 - c. $5 \times 10^4 \text{ J}$
 - d. $10 \times 10^4 \text{ J}$
8. The pressure and density of diatomic gas changes adiabatically from (p, d) to $(p', 32d)$ then value of $\frac{p'}{p}$ is
 - a. $\frac{1}{128}$
 - b. $\frac{2}{64}$
 - c. 64
 - d. 128
9. If the temperature of a perfectly black body increased by 25% then the emittance of a body increased by
 - a. 50%
 - b. 100%
 - c. 144%
 - d. 198%
10. When viewed a white light soap bubbles shows colors because of
 - a. Interference
 - b. diffraction
 - c. scattering
 - d. dispersion
11. An object is initially at a distance of 100 cm from a plane mirror. If the mirror approaches the object at a speed of 5 cm/sec , then after 6 s the distance between the object and its image will be
 - a. 60 cm
 - b. 140 cm
 - c. 170 cm
 - d. 150 cm .
12. A convex lens of focal length 12 cm is made of glass of $\mu_g = 3/2$. The focal length in liquid of refractive index $\mu_l = 5/4$ is
 - a. 6 cm
 - b. 12 cm
 - c. 24 cm
 - d. 30 cm
13. A lamp is hanging along the axis of circular table of radius R . At what height should the lamp be placed above the table so that illuminance at edge of table is $1/8$ of center?
 - a. $\frac{R}{\sqrt{3}}$
 - b. $\frac{2}{\sqrt{3}}$
 - c. $\sqrt{3}$
 - d. $\frac{\sqrt{3}}{2} R$
14. For a stationary wave $y = 45 \text{ m} \cdot \cos 96\pi t$ (x and y in cm). Find the distance between node and next antinode.
 - a. 6 cm
 - b. 7.5 cm
 - c. 9.5 cm
 - d. 4 cm

15. There are three sources of sound of equal intensities with frequency 200, 201, 202 vibrations /sec. The number of beat per second.
a. 0 b. 1 c. 2 d. 3
16. A beam of sound is 10^6 times as intense as that with minimum audible intensity. The intensity of the beam is then
a. 10^6 db b. 6 db c. 60 db d. 0.6 db
17. When 10^{14} electrons are removed from a neutral metal sphere, the charge on the sphere becomes
a. $16\mu\text{c}$ b. $-16\mu\text{c}$ c. $32\mu\text{c}$ d. $-32\mu\text{c}$
18. The ratio by which a charge Q should be divided so that they experience maximum force for a given separation is
a. 1:1 b. 1:2 c. 1:4 d. 12:3
19. The resistance of a wire is r ohm. The wire is stretched to double its length. Now the resistance of wire in ohm is
a. $\frac{r}{2}$ b. $4r$ c. $2r$ d. $\frac{r}{4}$
20. To send 10% of the main current through a moving coil galvanometer of resistance 99 ohm. The shunt required is
a. 9.9 ohm b) 100 ohm c) 11 ohm d) 19 ohm
21. 40 ohm electric heater is connected to a 200V, 50HZ main supply. The peak value of electric current flowing in the circuit is approximate.
a. 2.5 A b) 5.0 A c) 7A d) 10A
22. An ion for which the charge per unit mass is 4.4×10^7 c/kg has velocity 3.5×10^5 m/sec and moves in a circular path in a magnetic field 0.4T. The radius of circular path is.
a. 1 cm b) 2cm c) 3cm d) 4cm
23. A monochromatic source of light operating at 200W emits is 4×10^{20} photons per second, the wavelength of light is
a. 400 nm b) 200 nm c) is 4×10^{-10} nm d) none
24. The time period in initial state is eight times the period in final state. The ratio of the energy level is
a. 1:2 b) 1:3 c) 2:1 d) 4:3
25. The current gain alpha of a transistor is 0.95. The change in emitter current is 10 mA. The change in base current is
a. 9.5 mA b) 0.5 mA c) 10.5 mA d) (200/19) mA

English (25X1=25)

26. Government has a lot to road accidents.
(a) answer back (b) answer for (c) answer to (d) answer into
27. He inherited a fortune and it in a year.
(a) ran into (b) ran over (c) ran through (d) ran onto
28. His father blamed him but his mother and said he had acted sensibly.
(a) stand for (b) stand up to (c) stand out (d) stand up for
29. Do you me a fool?
(a) take, for (b) take, in (c) take, to (d) take, after
30. The captain , not realizing that the icebergs were so close.
(a) turn in (b) turn on (c) turn away (d) turn down
31. After a month's treatment at a hospital my brother is
(a) As good as good (b) as heavy as lead (c) As gay as a lark (d) as fit as fiddle
- Direction:** Select the word/phrase which is closest to the opposite or similar in meaning of the word.
32. Squeamish (similar meaning)
(a) very dirty (b) gullible (c) easily upset (d) braggart
33. Bramble (similar meaning)

- (a) wild cat (b) wild bush with thorn (c) tall plant (d) wild animal
34. Congruous (similar meaning)
(a) consistent (b) covert (c) Irregular (d) clamorous
35. savior (opposite meaning)
(a) protector (b) destroyer (c) nurturer (d) designer
36. Scabrous (opposite meaning)
(a) scaly (b) scabby (c) decent (d) zigzag
37. un-availing (opposite meaning)
(a) unavailable (b) re past (c) successful (d) rare
- Direction:** Select the proper prepositions and conjunctions to fill in the blanks in the following questions:
38. He died cancer but she died a train accident
(a) out, at (b) off, by (c) for, in (d) of, in
39. Divide forty Eight but divide the these apples the two children.
(a) for, among (b) at, between (c) in, among (d) by, between
40. It is a lovely place to live
(a) in (b) for (c) at (d) by
41. We generally add”s” to make the nouns plural.
(a) a (b) an (c) nothing (d) the
42.are to be blamed soon.
(a) He,you and I (b) I, he and you (c) you, he and I (d) He, I and you
43. He is angry and indifferentme.
(a) from, from (b) by, by (c) with, to (d) during, by
44. You alone can relieve me This anxiety.
(a) from (b) of (c) in (d) during
45. He took the food play with your watch he breaks it.
(a) should (b) in case (c) lest (d) might
- Direction:** Each of the following idioms an phrases is followed by four explanations. Mark the one that best explains its meaning.
46. On the bottle
(a) teetotaler (b) vegetarian (c) drinking a lot (d) avoid alcohol
47. On the rocks
(a) in great success (b) in serious trouble (c) heavy star (b) in progress
48. Call it a night
(a) stop some activity (b) feel lazy (c) enjoy the night (d) belated activity
49. Bide one's time
(a) to be punctual (b) swearing frequently (c) use rough language (d) act like a soldier
50. Over the hill
(a) high land (b) mountain area (c) most difficult part (d) rough road

Mathematics (25X1=25)

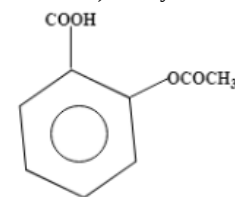
51. If $A \cap X = A \cap Y$ and $A \cup X = A \cup Y$ then, $x =$
a) $A \cap X$ b) $A \cap Y$ c) Y d) $Y \cap X$
52. If $f(x) = \frac{x}{x-1} = \frac{1}{y}$, then $f(y) =$
a) X b) $x + 1$ c) $x-1$ d) $1-x$
53. If $\log_a \sqrt{3} = \frac{1}{2}$, then the value of a is:
a) 0 b) 1 c) 3 d) $\frac{1}{2}$

54. If $\begin{vmatrix} -a^2 & ab & ac \\ ab & -b^2 & bc \\ ac & bc & -c^2 \end{vmatrix} = ka^2b^2c^2$, then the value of k is
 a) 1 b) -2 c) 4 d) -4
55. The domain of the function $f(x) = \sqrt{|x| - x}$ is
 a) $(0, \infty)$ b) $(-\infty, 0]$ c) $(-\infty, \infty)$ d) $(-1, 1)$
56. If the difference of the roots of the equation $x^2 + kx + 7 = 0$ is 6, then the possible values of k are
 a) 4 b) -4 c) -8 d) ± 8
57. If R is the H.M between P & Q, then $\frac{R}{P} + \frac{R}{Q} =$
 a) $\frac{PQ}{P+Q}$ b) $\frac{P+Q}{PQ}$ c) 2 d) none
58. If w is the cube root of unit, then the value of $(1 - w + w^2)^5 + (1 + w - w^2)^5 =$
 a) 16 b) 32 c) 48 d) -16
59. The value of $(i + i)^{20}$ is
 a) 2^{10} b) 2^{15} c) 2^{20} d) -2^{10}
60. For what value of k the roots of the equation $3x^2 - 2kx + k = 0$ are the ratio 3:1
 a) 0 or 4 b) 0 or -4 c) 1 or 3 d) 0 or 1
61. The value of $\tan\left(\frac{\pi}{4} - \tan^{-1}\frac{2}{3}\right)$ is
 a) $\frac{1}{5}$ b) $\frac{7}{17}$ c) $\frac{-3}{4}$ d) $\frac{1}{2}$
62. $\tan\theta + \tan 2\theta + \tan\theta \tan 2\theta = 1$ has general solution equations
 a) $(2n+1)\frac{\pi}{6}$ b) $2n\pi \pm \frac{\pi}{6}$ c) $n\pi \pm \frac{\pi}{4}$ d) $(4n+1)\frac{\pi}{12}$
63. The value of the h if the equation $4x^2 + hxy + y^2 = 0$ represents the coincident lines is:
 a) ± 3 b) ± 5 c) ± 4 d) ± 2
64. The radius of the circle $x^2 + y^2 - 2x\cos\theta + 2y\sin\theta - 8 = 0$ is:
 a) 8 b) $2\sqrt{3}$ c) $\sqrt{3}$ d) 3
65. The polar coordinates of $(2, 2\sqrt{3})$ is
 a) $(2, 30^\circ)$ b) $(4, 60^\circ)$ c) $(2, 55^\circ)$ d) $(2, 90^\circ)$
66. The Cartesian coordinates (x, y) of $(2, \frac{\pi}{3})$ is
 a) $(2\sqrt{2}, 2\sqrt{2})$ b) $(1, 2)$ c) $\sqrt{3}, \sqrt{2}$ d) $(1, \sqrt{3})$
67. The general equation of 2nd degree $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a circle if:
 a) $a + b = 0$ b) $a = b = h = 0$ c) $a = b$; coeff of $xy = 0$ d) None
68. $\frac{d^2y}{dx^2} = 0$, then
 a) $y = ax + b$ b) $y^2 = ax + b$ c) $y = \log x$ d) $y = e^2 + c$
69. $\lim_{x \rightarrow 0} \frac{1 - \cos px}{1 - \cos mx}$
 a) $\frac{p}{m}$ b) $\frac{p^2}{m^2}$ c) $\frac{m}{p}$ d) $\frac{m^2}{p^2}$
70. If $f(x)$ be a quadratic polynomial such that $f(0) = 2$, $f'(0) = -3$ and $f''(0) = 4$ then $f(x) =$
 a) $3x^2 - 4x + 2 = 0$ b) $2x^2 - 3x + 2 = 0$ c) $x^2 - 3x + 2 = 0$ d) None
71. Area bounded by the curve $y = x^2$ and $y = 2 - x^2$ is
 a) $\frac{8}{3}$ b) $\frac{3}{8}$ c) $\frac{3}{2}$ d) $\frac{1}{3}$
72. The derivation of $e^x \sin x$ w.r.t. e^x is:
 a) $\sin x + \cos x$ b) $e^x(\sin x + \cos x)$ c) $\sin x$ d) $e^x \cos x$
73. $\int_{-1}^1 |1 - x| dx =$
 a) -2 b) 0 c) 2 d) 4
74. Every continuous function is

- a) Increasing b) decreasing c) not always differentiable d) always differentiable
75. If $e^{\sin x} + e^{\sin y} = 1$ then $\frac{dy}{dx} =$
 a) $\frac{e^{\sin x}}{e^{\sin y}}$ b) $-\frac{(e^{\sin x} \cos x)}{e^{\sin y} \cos y}$ c) $\frac{\sin y \cdot e^{\sin x}}{\cos y \cdot e^{\cos x}}$ d) None

Chemistry (25X1=25)

76. 1 mole of O_2 equal to
 a) $2 \times 6.023 \times 10^{23} \times 8$ Moleculs b) 22.4 lit. at NTP
 c) 11.2 lit at NTP d) All of above
77. Carbon monoxide is
 a) Neutral oxide b) acidic oxide c) Basic oxide d) Amphoteric oxide
78. The structure of CO_2 molecule is
 a) Linear b) Tetrahedral c) Angular d) Pyramidal
79. Downs process is used for extraction of
 a) Sodium b) sodium hydroxide c) Sodium carbonate d) Ammonia
80. "Atomic orbitals are filled up in the ground state in order of increasing energy level" This statement is called
 a) Pauli's exclusion principle b) Hund's rules
 c) Aufbau Principle d) Heisenberg uncertainty principle
81. Gibb's Helmholtz equation is
 a) $\Delta E = Q + W$ b) $\Delta G = \Delta H - T\Delta S$ c) $G = H + TS$ d) $\Delta E = Q - W$
82. The pH of $10^{-2} N H_2SO_4$ is
 a) 3 b) 4 c) 2.5 d) 2
83. Which of the following compound is not a primary standard substance
 a) Oxalic acid b) Potassium dichromate c) Potassium bromate d) Potassium Permanganate
84. The reaction of benzene with alkyl chloride in presence of anhydrous $AlCl_3$ is
 a) Friedel-Craft Reaction b) Wurtz reaction c) Reimer-Tiemann's reaction d) Grignard reaction
85. The member of homologous series have
 a) Different general formula b) different molecular weight
 c) Different chemical properties d) different method of properties
86. Which of the following reagent cannot be used to distinguish between acetaldehyde and acetone
 a) Tollens's reagent b) Fehling's solution c) Schiff's reagent d) Br_2 in CCl_4
87. Reduction of nitrobenzene in acidic medium gives
 a) Aniline b) Azoxybenzene c) ozobenzene d) hydrazobenzene



88. The compound
 a) Antibiotic b) antiseptic c) analgesic d) pesticide
89. Which of the following of compounds forms zwitter ion?
 a) carbohydrates b) alpha amino acids c) nucleic acid d) lipids
90. Which of the following undergoes nitration easily?
 a) Toluene b) Benzene c) carboxylic acid d) nitrobenzene
91. BHC is an example
 a) Condensation polymer b) addition polymer c) azodye d) pesticide
92. Orthophosphoric acid

- | | | | |
|-------------------|-----------------|------------------|--------------------|
| a) monobasic acid | b) dibasic acid | c) Tribasic acid | d) Tetrabasic acid |
|-------------------|-----------------|------------------|--------------------|
93. Baeyer's process is used for manufacture of
- | | | | |
|-------------------|----------------|------------|-------------|
| a) Sulphuric acid | b) Nitric acid | c) Ammonia | d) Aluminum |
|-------------------|----------------|------------|-------------|
94. The formula of malachite green is
- | | | | |
|----------------------------|-----------------------------|-------------------------|-------------|
| a) $Cu(OH)_2 \cdot CuCO_3$ | b) $2CuCO_3 \cdot Cu(OH)_2$ | c) $ZnSO_4 \cdot 7H_2O$ | d) $HgCl_2$ |
|----------------------------|-----------------------------|-------------------------|-------------|
95. The general electronic configuration of Cu is
- | | | | |
|-------------------|-------------------|-------------------|----------------------|
| a) $[Ar]4s^13d^5$ | b) $[Ar]4s^23d^8$ | c) $[Ar]4s^23d^9$ | d) $[Ar]4s^13d^{10}$ |
|-------------------|-------------------|-------------------|----------------------|
96. German silver is alloy of
- | | | | |
|------------------|------------------|--------------|--------------|
| a) Cu, Sn and Zn | b) Cu, Zn and Ni | c) Cu and Sn | d) Cu and Zn |
|------------------|------------------|--------------|--------------|
97. Which of the following is the most metallic?
- | | | | |
|-------|-------|------|-------|
| a) Na | b) Al | c) K | d) Zn |
|-------|-------|------|-------|
98. Number of atom in molecule of Sulphur is
- | | | | |
|------|------|------|------|
| a) 8 | b) 4 | c) 2 | d) 1 |
|------|------|------|------|
99. Chalcocopyrite is an ore of
- | | | | |
|-------|-------|-------|-------|
| a) Au | b) Cu | c) Zn | d) Hg |
|-------|-------|-------|-------|
100. Heavy water is
- | | | | |
|-----------|-----------|-------------|-----------|
| a) H_2O | b) D_2O | c) H_3O^+ | d) OH^- |
|-----------|-----------|-------------|-----------|

--Best of Luck --

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